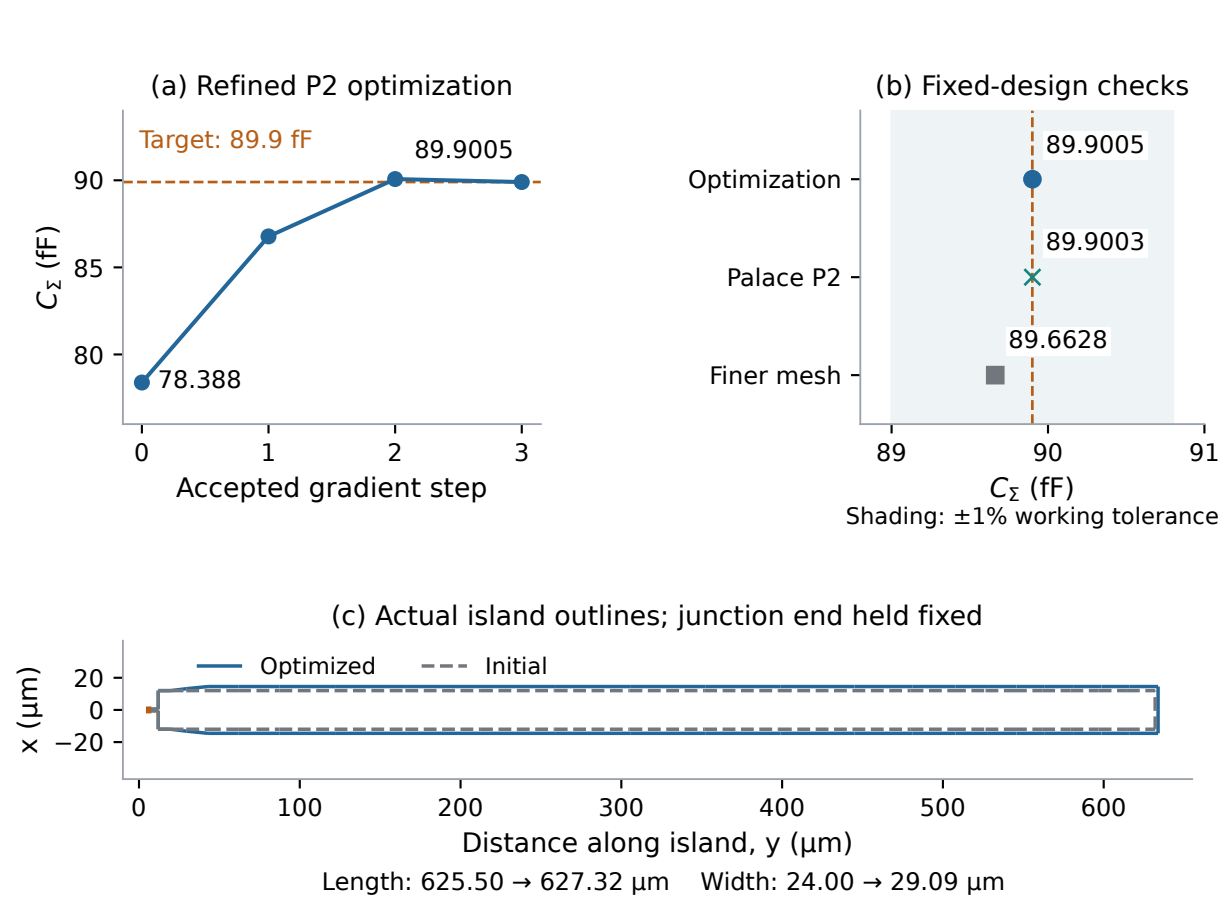
(a) Refined P2 optimization
Target: 89.9 fF
89.9005
78.388
C_Σ (fF)
90
85
80
0
1
2
3
Accepted gradient step
(b) Fixed-design checks
89.9005
Optimization
89.9003
Palace P2
89.6628
Finer mesh
89
90
91
C_Σ (fF)
Shading: ±1% working tolerance
(c) Actual island outlines; junction end held fixed
Optimized
Initial
x (µm)
20
0
−20
0
100
200
300
400
500
600
Distance along island, y (µm)
Length: 625.50 → 627.32 µm
Width: 24.00 → 29.09 µm